Evaluating adaptation to changing networks beyond accuracy in a fixed environment

Updating IDS models with few labeled examples at low cost using pretrained TabPFN

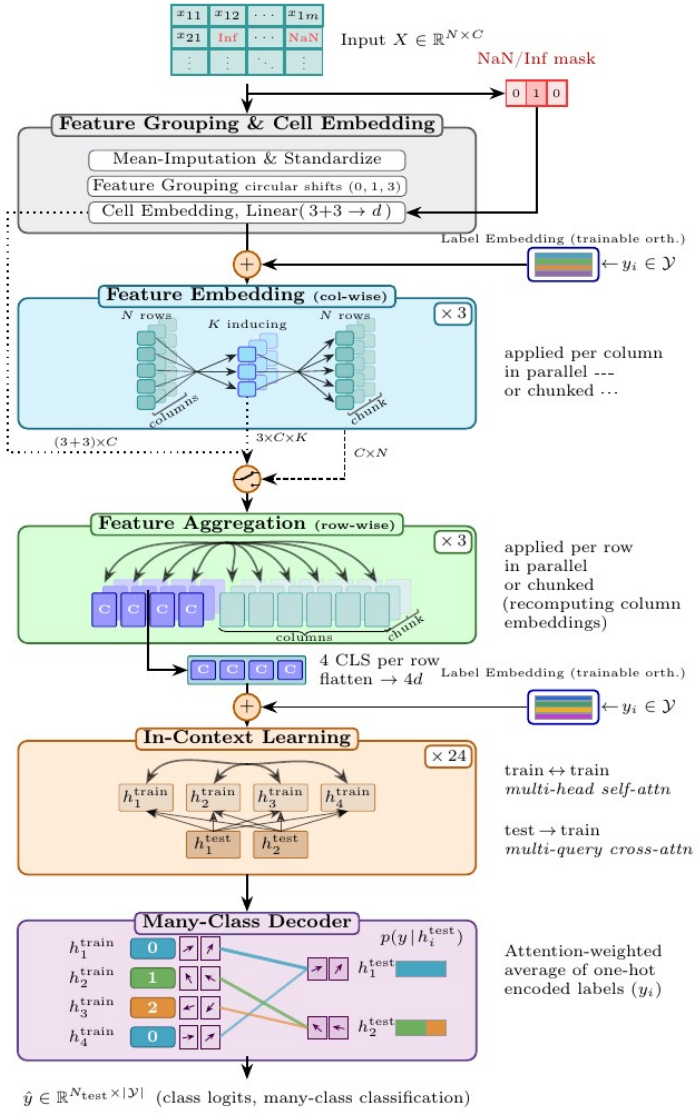
TabPFN · Pretrained prediction rules for tabular data (v3: arXiv, 2026)

Pretraining	Pretrained on over 8 trillion tokens of synthetic tables Learns prediction rules by predicting hidden targets
In-context learning	Predicts a new flow using labeled examples as context Model weights remain fixed during inference
Fine-tuning	Splits initial IDS data into context and query examples Backpropagates query loss to adapt the model weights

Update: fixed pretrained weights → retain old classes and add new examples

Hypothesis: comparable detection quality with fewer labels and lower update cost

v3 architecture



① Encode inputs · Cell embedding

Converts values such as packet length into vectors
Also indicates which entries are missing

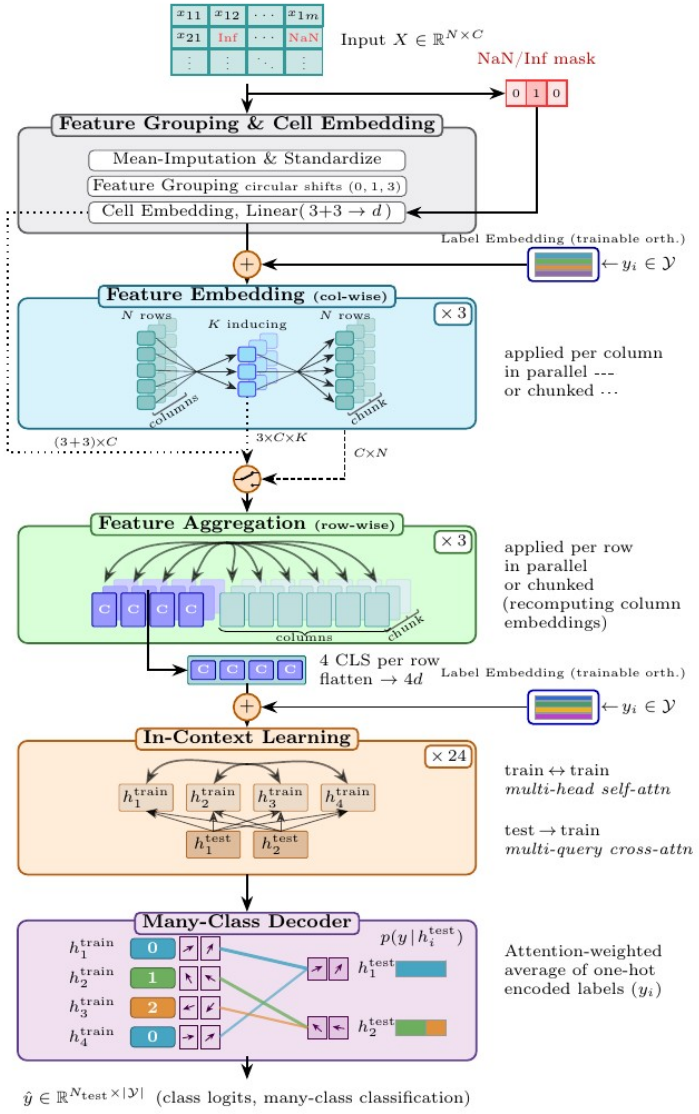
② Summarize columns · Feature embedding

Summarizes each feature across multiple flows
Uses the benign and attack labels in the context

③ Combine features · Feature aggregation

Combines each flow's features into one representation
Uses it to compare relationships between flows

v3 architecture



④ Consult examples · In-context learning

A new flow queries the labeled context examples

Attention identifies relevant examples for prediction

⑤ Predict probabilities · Many-class decoder

Assigns greater weight to labels of relevant examples

Combines them into benign and attack probabilities

Keep context examples of existing classes
to continue predicting those classes

Evaluation scenario

Initial 1,000 · Add 100 per new attack class · Stage order, then descending sample count

CIC2018

Stage	New classes	Per class	Cumulative
1	Benign Brute force DoS	750 125 125	1,000
2	DDoS	100	1,100
3	Web attacks	100	1,200
4	Infiltration	100	1,300
5	Bot	100	1,400

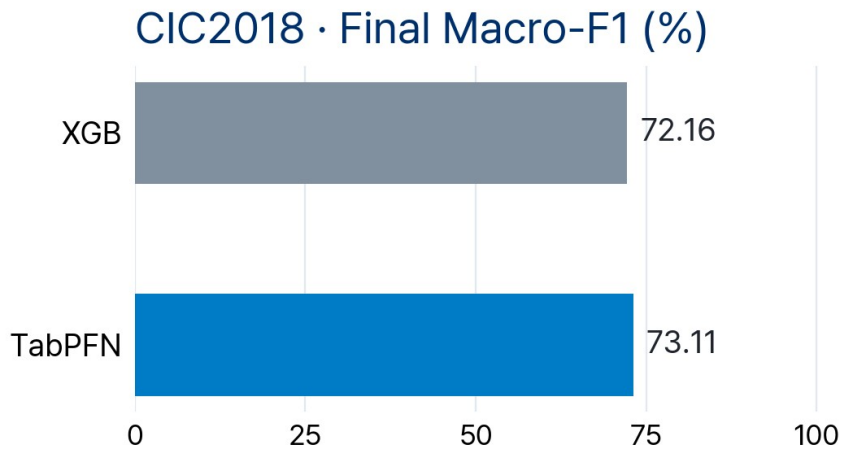
ToN

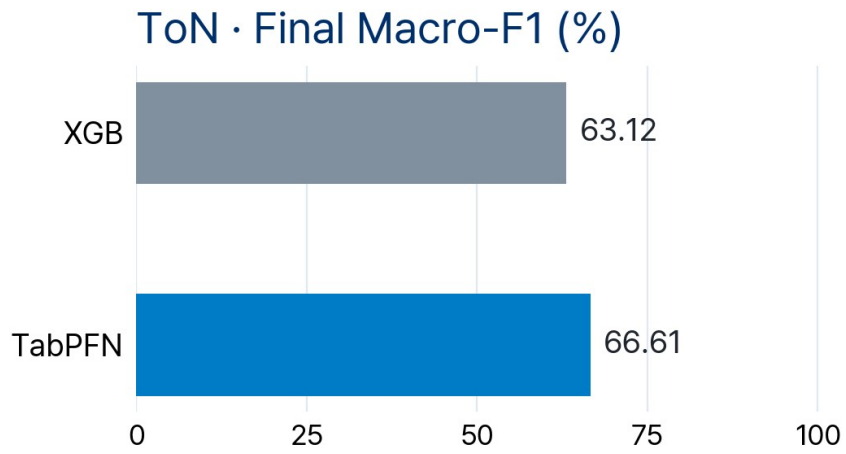
Stage	New classes	Per class	Cumulative
1	Benign DoS Scanning	750 125 125	1,000
2	DDoS Injection	100 100	1,200
3	Password	100	1,300
4	XSS	100	1,400
5	Backdoor Ransomware	100 100	1,600
6	MitM	100	1,700

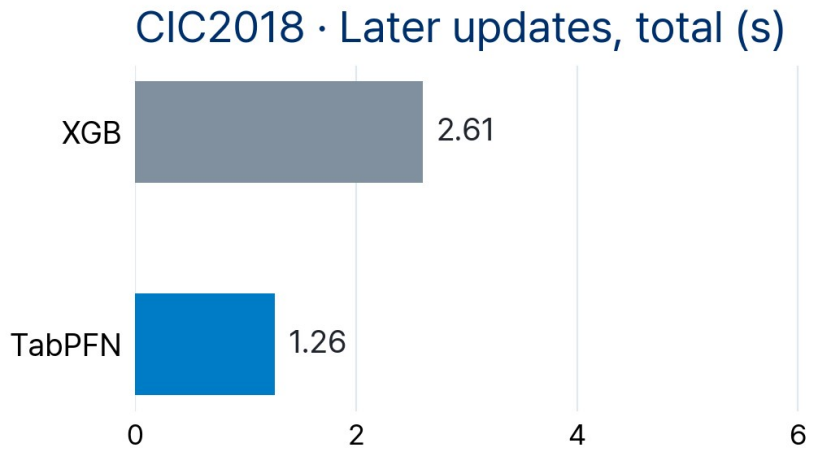
Full test: CIC2018 3,042,473 / ToN 2,271,723 · Final model · Mixed order, seed 43

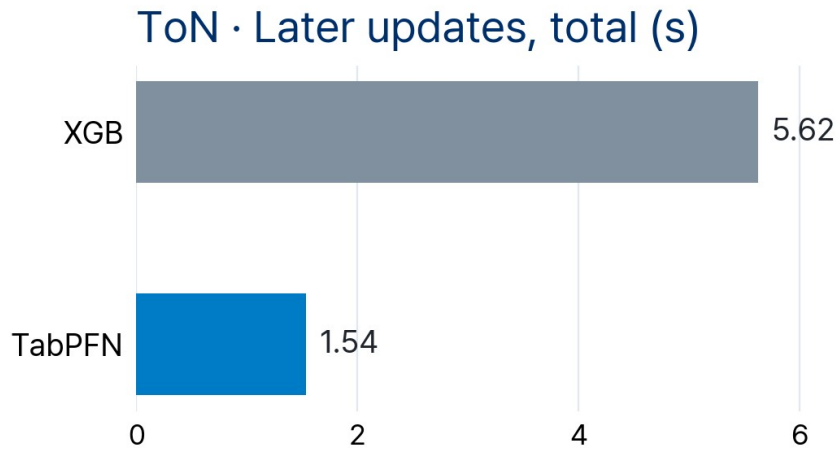
Results

Final model, full test · XGB: cumulative retraining · TabPFN: cumulative context, no FT







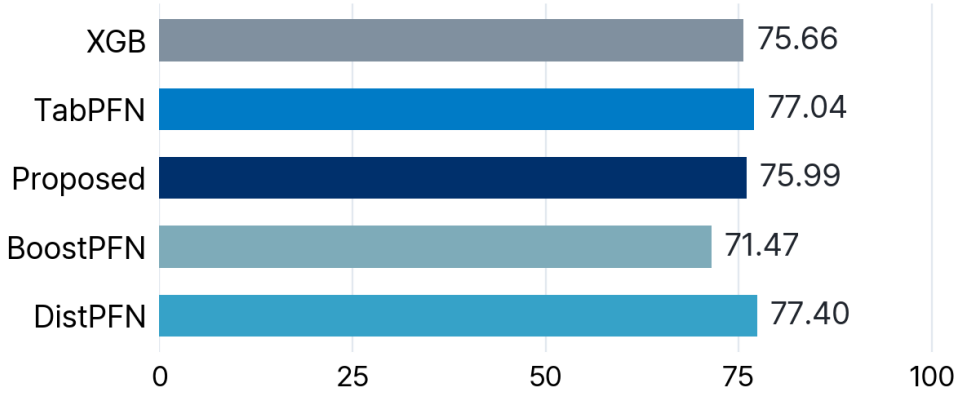


Cost: updates from stage 2 onward · Initial setup and inference excluded · Shared time scale

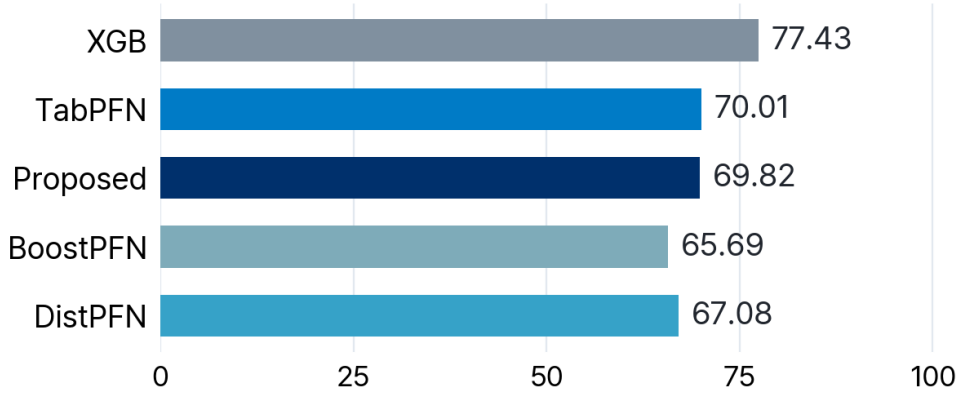
100k training samples · Proposed model and SOTA

75,000 benign + 25,000 attack samples · Final model, full test · Macro-F1 (%)

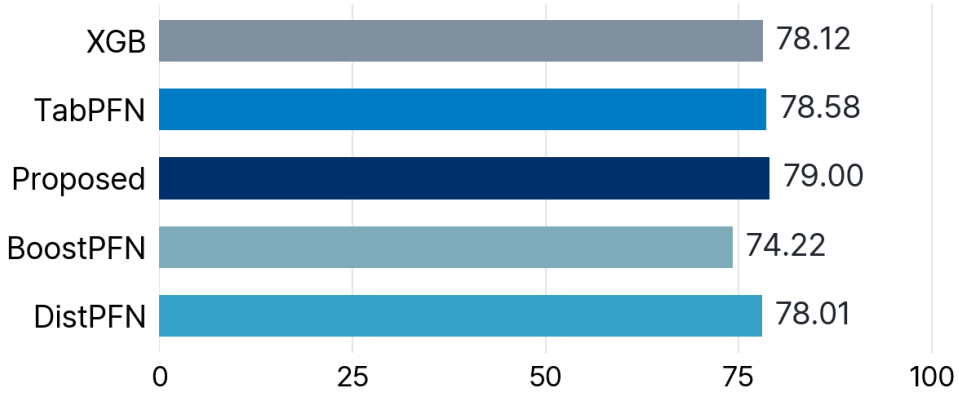
CIC2018 · Natural proportions



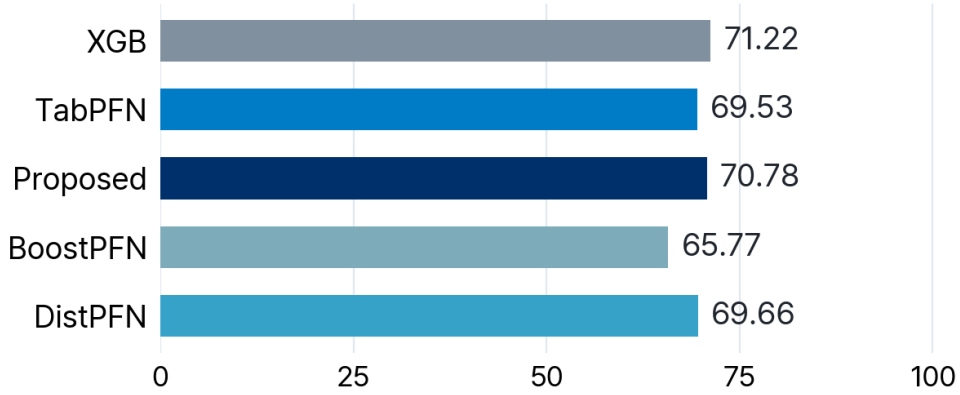
ToN · Natural proportions



CIC2018 · Uniform allocation



ToN · Uniform allocation

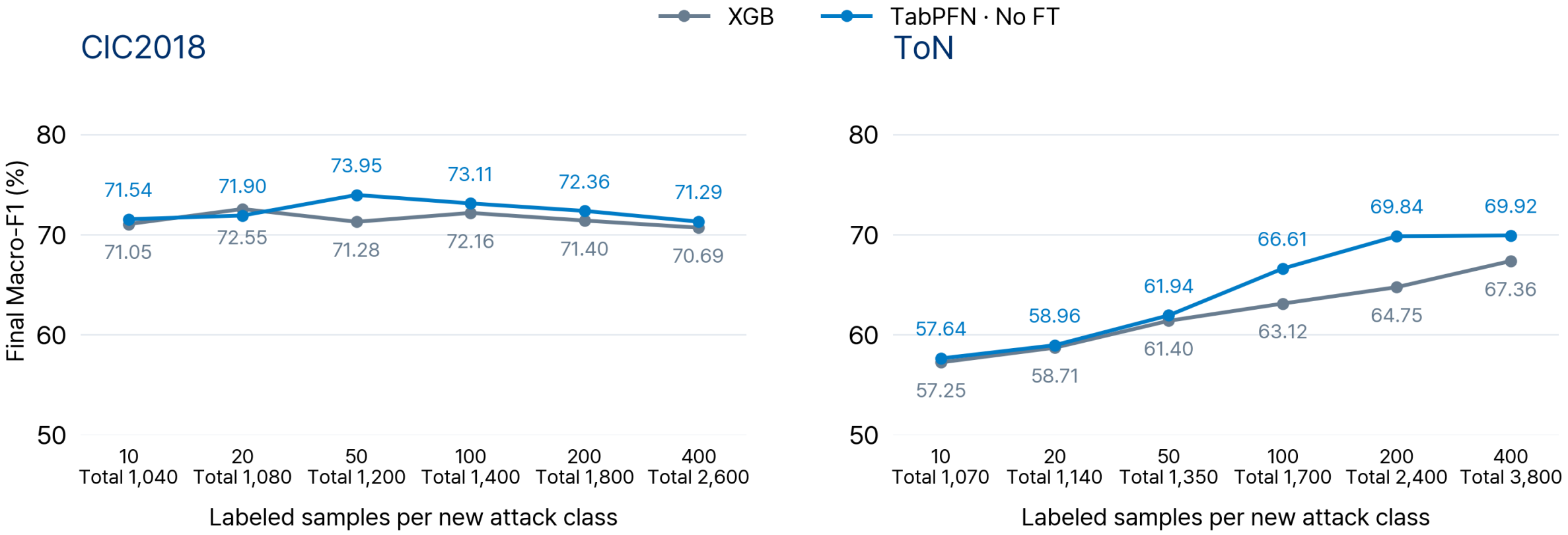


Proposed: Global ≈50k + separate validation · Counts: pp. 9, 10 · BoostPFN v1, other PFNs v3

Proposed: 79.00% on CIC2018 uniform · XGB leads both ToN allocations

Performance by new attack sample budget

Initial 1,000 fixed · 10/20/50/100/200/400 per new attack class · Final model, full test · Seed 43



400 per new attack · TabPFN minus XGB: CIC2018 +0.60 pp · ToN +2.56 pp

Appendix · 100k training sample scenario · CIC2018

Same attack order · Small: 100 per new attack · Large: 15,000 new benign samples per stage

		New samples at this stage			Cumulative samples		
Stage	**Added classes**	**Small budget**	**100k natural**	**100k uniform**	**Small budget**	**100k natural**	**100k uniform**
1	Benign DoS Brute force	750 125 125	15,000 2,518 2,403	15,000 4,910 4,910	1,000	19,921	24,820
2	DDoS Benign	100 0	16,780 15,000	4,910 15,000	1,100	51,701	44,730
3	Benign Web attacks	0 100	15,000 9	15,000 450	1,200	66,710	60,180
4	Benign Infiltration	0 100	15,000 641	15,000 4,910	1,300	82,351	80,090
5	Benign Bot	0 100	15,000 2,649	15,000 4,910	1,400	100,000	100,000
	Separate val.				0	8,211	8,047

Separate validation: residual calibration and expert acceptance · Small setting: within total budget

Natural: training attack ratios · Uniform: equal attack quotas, availability caps · Sorted by stage, then natural additions

Appendix · 100k training sample scenario · ToN

Same attack order · Small: 100 per new attack · Large: 12,500 new benign samples per stage

		New samples at this stage			Cumulative samples		
Stage	**Added classes**	**Small budget**	**100k natural**	**100k uniform**	**Small budget**	**100k natural**	**100k uniform**
1	Benign DoS Scanning	750 125 125	12,500 644 168	12,500 2,856 2,856	1,000	13,312	18,212
2	Benign DDoS Injection	0 100 100	12,500 5,785 1,899	12,500 2,856 2,856	1,200	33,496	36,424
3	Benign Password	0 100	12,500 6,009	12,500 2,856	1,300	52,005	51,780
4	Benign XSS	0 100	12,500 9,308	12,500 2,856	1,400	73,813	67,136
5	Benign Backdoor Ransomware	0 100 100	12,500 1,136 20	12,500 2,857 2,151	1,600	87,469	84,644
6	Benign MitM	0 100	12,500 31	12,500 2,856	1,700	100,000	100,000
	Separate val.				0	8,668	8,605

Separate validation: residual calibration and expert acceptance · Small setting: within total budget

Natural: training attack ratios · Uniform: equal attack quotas, availability caps · Sorted by stage, then natural additions